

Q1.

The linear transformations T_1 and T_2 are represented by the matrices

$$\mathbf{M}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_2 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

respectively.

(a) Give a full geometrical description of the transformations:

(i) T_1 ;

(2)

(ii) T_2 .

(3)

(b) Find the matrix which represents the transformation T_1 followed by T_2 .

(2)

(c) The linear transformation T_3 is represented by the matrix

$$\mathbf{M}_3 = \begin{bmatrix} k & 2 & -1 \\ 1 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}$$

where k is a constant.

For one particular value of k , T_3 has a line L of invariant points.

(i) Find k .

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(ii) Find the Cartesian equations of L in the form $\frac{x}{p} = \frac{y}{q} = \frac{z}{r}$.

(7)

(Total 14 marks)

Q2.

A line and a plane have equations

$$\frac{x-3}{p} = \frac{y-q}{3} = \frac{z-1}{-1}$$

and

$$\mathbf{r} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 10$$

respectively, where p and q are constants.

(a) Show that the line is **not** perpendicular to the plane. (1)

(b) In the case where the line lies in the plane, find the values of p and q . (4)

(c) In the case where the angle, θ , between the line and the plane satisfies $\sin \theta = \frac{1}{\sqrt{6}}$, and the line intersects the plane at $z = 2$:

(i) find the value of p ; (5)

(ii) find the value of q . (2)

(Total 12 marks)

Q3.

The planes Π_1 and Π_2 have equations

$$\mathbf{r} \cdot \begin{bmatrix} 2 \\ 1 \\ 7 \end{bmatrix} = 10 \quad \text{and} \quad \mathbf{r} \cdot \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix} = 7$$

respectively.

(a) Determine, to the nearest degree, the acute angle between Π_1 and Π_2 . (4)

(b) By setting $z = t$, find cartesian equations for the line of intersection of Π_1 and Π_2 in the form

$$\frac{x-a}{l} = \frac{y-b}{m} = z=t$$

(6)

(c) The line L , with equation $\mathbf{r} = \begin{bmatrix} 20 \\ -1 \\ 7 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 9 \\ 4 \end{bmatrix}$, intersects Π_1 at the point P and Π_2 at the point Q .

Show that $PQ = k\sqrt{2}$, where k is an integer.

(6)

(Total 16 marks)

Q4.

$$\begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix} = 11$$

The plane Π has equation $\mathbf{r} \cdot \begin{bmatrix} 12 \\ 15 \\ 16 \end{bmatrix} = 11$ and the point Q has coordinates $(1, 1, -1)$.

- (a) Show that Q is in Π . (1)
 - (b) Write down cartesian equations for the line l which passes through Q and is perpendicular to Π . (2)
 - (c) The points M and N are on l , and each is 50 units from Π .
Find the coordinates of M and N . (3)
 - (d) Given that the point $P(5, 1, -4)$ is in Π , determine the area of triangle PMN . (3)
- (Total 9 marks)**

Q5.

- (a) The transformation U of three-dimensional space is represented by the matrix

$$\begin{bmatrix} 1 & 4 & -3 \\ 2 & -1 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

- (i) Write down a vector equation for the line L with cartesian equation

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{6}$$

- (ii) Find a vector equation for the image of L under U , and deduce that it is a line through the origin. (2)
- (ii) Find a vector equation for the image of L under U , and deduce that it is a line through the origin. (4)

- (b) The plane transformation V is represented by the matrix $\begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix}$.

L_1 is the line with equation $y = \frac{1}{2}x + k$ and L_2 is the image of L_1 under V .

- (i) Find, in the form $y = mx + c$, the cartesian equation for L_2 . (4)
- (ii) Deduce that L_2 is parallel to L_1 and find, in terms of k , the distance between these two lines. (3)

Q6.

(a) The line l has equation $\mathbf{r} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 3 \\ 2 \\ 6 \end{bmatrix}$.

- (i) Write down a vector equation for l in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$.

(1)

- (ii) Write down cartesian equations for l .

(2)

(b) The plane Π has equation $\mathbf{r} = \begin{bmatrix} 7 \\ 5 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 4 \\ 3 \\ 2 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$.

- (i) Find an equation for Π in the form $\mathbf{r} \cdot \mathbf{n} = d$.

(4)

- (ii) State the geometrical significance of the value of d in this case.

(1)

- (c) Determine, to the nearest 0.1° , the angle between l and Π .

(4)

(Total 12 marks)

Q7.

The coordinates of the points A and B are $(3, -2, 1)$ and $(5, 3, 0)$ respectively.

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The line l has equation $\mathbf{r} = \begin{bmatrix} 5 \\ 3 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$.

- (a) Find the distance between A and B .

(2)

- (b) Find the acute angle between the lines AB and l . Give your answer to the nearest degree.

(5)

- (c) The points B and C lie on l such that the distance AC is equal to the distance AB . Find the coordinates of C .

(5)

(Total 12 marks)

Q8.

- (a) Find, to the nearest 0.1° , the acute angle between the planes with equations

$$\mathbf{r} \cdot (3\mathbf{i} - 4\mathbf{j} + \mathbf{k}) = 2 \quad \text{and} \quad \mathbf{r} \cdot (2\mathbf{i} + 12\mathbf{j} - \mathbf{k}) = 38$$

(4)

- (b) Write down cartesian equations for these two planes.

(2)

- (c) Find, in the form $\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n}$, cartesian equations for the line of intersection of the two planes.

(5)

(Total 11 marks)

Q9.

Find cartesian equations for the line with vector equation

$$\mathbf{r} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} -2 \\ 6 \\ 3 \end{bmatrix}$$

giving your answer in the form $\frac{x-a_1}{b_1} = \frac{y-a_2}{b_2} = \frac{z-a_3}{b_3}$

(Total 3 marks)